



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT/2024-25/LMI/EMHT/13363

Date:04/04/2024(16:40:51)

Subject:REVIEW OF LMI ON HYDROGEN PLANT

Sub: REVIEW OF LMI ON HYDROGEN PLANT (LMI/SMPP/OPS/ELEC/026)

LMIs are reviewed periodically for further improvement and incorporation of recommendations by internal or external audit committee and COS.This Local Management instruction formulated locally i.e at plant level.No changes envisaged.

Submitted for review & approval of competent authority please.

Submitted by:

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On:04/04/2024(16:40:51) **Ref:**SMTTP/ELECT MAINT/2024-25/LMI/EMHT/13363

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On:10/04/2024(17:31:14) **Ref:**SMTTP/ELECT MAINT/2024-25/LMI/EMHT/13363

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On:12/04/2024(11:52:44) **Ref:**SMTTP/ELECT MAINT/2024-25/LMI/EMHT/13363

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On:12/04/2024(18:10:41) **Ref:**SMTTP/ELECT MAINT/2024-25/LMI/EMHT/13363

Note No:5

MAY BE APPROVED

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On:13/04/2024(11:19:35) **Ref:**SMTTPP/ELECT MAINT/2024-25/LMI/EMHT/13363

Note No:6

May pl review

Submitted by:

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On:14/04/2024(22:19:46) **Ref:**SMTTPP/ELECT MAINT/2024-25/LMI/EMHT/13363

Note No:7

C&I responsibilities reviewed.

Submitted by:

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On:15/04/2024(10:33:24) **Ref:**SMTPP/ELECT MAINT/2024-25/LMI/EMHT/13363

Note No:8

May be approved

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Note No:9

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On:15/04/2024(11:15:28) **Ref:**SMTTP/ELECT MAINT/2024-25/LMI/EMHT/13363

Note No:10

Pl review

Submitted by:

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On:15/04/2024(15:54:06) **Ref:**SMTTP/ELECT MAINT/2024-25/LMI/EMHT/13363

Note No:11

Point no 6.0/C may please be reviewed.

Submitted by:

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On:15/04/2024(18:38:33) **Ref:**SMTPP/ELECT MAINT/2024-25/LMI/EMHT/13363

Note No:12

Sir,point No: 6.0/C reviewed and corrected.

Submitted by:

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On:15/04/2024(18:45:15) **Ref:**SMTPP/ELECT MAINT/2024-25/LMI/EMHT/13363

Note No:13

May please be approved.

Submitted by:

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On:15/04/2024(18:48:02) **Ref:**SMTPP/ELECT MAINT/2024-25/LMI/EMHT/13363

Note No:14

Submitted by:

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On:20/04/2024(09:06:05) **Ref:**SMTPP/ELECT MAINT/2024-25/LMI/EMHT/13363

Note No:15

Approved & Submitted by:

संजय कुमार सिन्हा - मुख्य महाप्रबंधक

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On:22/04/2024(13:06:10) **Ref:**SMTPP/ELECT MAINT/2024-25/LMI/EMHT/13363

NTPC LTD

SIMHADRI THERMAL POWER PLANT

LOCAL MANAGEMENT INSTRUCTION

LMI/SMPP/OPS/ELEC/026

ISSUE NO.: 3, REVISION-0, DATE: APR 2024

OPERATION AND MAINTENANCE OF HYDROGEN PLANT

APPROVED FOR

IMPLEMENTATION BY : _____

HOP (SIMHADRI)

DATE : _____

NTPC LTD**SIMHADRI THERMAL POWER PLANT****OPERATION & MAINTENANCE DEPARTMENT**

LMI(NAME) : OPERATION AND MAINTENANCE OF HYDROGEN PLANT		
LMI NO : LMI/SMPP/OPS/ELEC/026		
ISSUE NO : 3	REVISION : 0	DATE: APR 2024

	Designation	Signature/Name
Prepared by	Sr. Manager (EM)	
Checked by	DGM(EM)	
Reviewed by	AGM(EM)	
Reviewed by	AGM(C&I)	
Reviewed by	AGM(Chemistry)	
Reviewed by	GM(O)	
Reviewed by	GM(O&M)	
Approved by	HOP(Simhadri)	

OPERATION AND MAINTENANCE OF HYDROGEN PLANT**CONTENTS**

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OPERATION AND MAINTENANCE OF HYDROGEN PLANT

1.0 INTRODUCTION:

The Hydrogen plant of Simhadri has a capacity to generate 7.5 cubic meters of hydrogen per hour. The hydrogen generation plant is not in operation.

Station requirement is being met by purchase of H2 Cylinder Refilling through Purchase order by outside authorised agency. We are having licence for storing 760 Cylinders

2.0 SCOPE:

This LMI covers all the various process of the Hydrogen storage at hydrogen plant area

3.0 SUPERSEDED DOCUMENT:

LMI/SMPP/OPS/ELEC/026 ISSUE3 : Rev 0 dated 15.12.2019

4.0 RECORDS OF REVISION:

NIL

5.0 REFERENCE DOCUMENTS

LMI/SMPP/OPS/ELEC/026, Issue -03, Dt: Feb 2022

6.0 PROCEDURE TO BE FOLLOWED FOR SAFE HANDLING OF CYLINDER

- a) Hydrogen storage area is highly inflammable and for safety purpose smoking and HOT work strictly prohibited.
- b) Storage area should be maintained free from any combustible items.
- c) In the event of power failure no matchstick or open fire is allowed for lighting
- d) While transporting of cylinders, the cylinders should be handled carefully and valve caps to be ensured. No cylinder should be rolled/pushed/pulled on floor.

7.0 SPECIAL INSTRUCTIONS & RESPONSIBILITY CENTERS:

Responsibilities of O&M-Electrical Maintenance:

To arrange and coordinate the hydrogen cylinder loading and unloading at H2 plant store and to coordinate with C&I and Chemistry department for leak detection and purity checking.

Responsibilities of O&M-Chemistry:

On information of receipt of filled H₂ cylinders, checking of H₂ purity of each refilled cylinder received and the purity readings are to be entered in the purity register.

Responsibilities of O&M-C&I:

Monthly checking of hydrogen leak detector is to be done as per attached format

8.0 RECORDING KEEPING:

1. Records will be maintained by concerned area maintenance group in respective files

9.0 RESPONSIBILITY:

1. It shall be the responsibility of the respective Electrical maintenance group to ensure the hydrogen plant storage to be carried out as per the instructions /procedure contained in the LMI.
2. Overall responsibility lies with HOD (Electrical Maintenance).

10.0 REVIEW:

Documents will be reviewed once in two years by HOP (Simhadri).

Annexture-1**HYDROGEN CYLINDERS PURITY TEST REPORT**

Test date:

Sl.No.	CYLINDER No.	PURITY %
1		

O&M-Chemistry

O&M-EM

SIGNATURE:

NAME :

DATE :

DEPARTMENT: O&M-C&I,EMD				SIMHADRI
Doc No. C&I/FRM - 02		DOCUMENT TITLE: FORMAT FOR HYDROGEN PLANT ISO INSTRUMENTS		REVISION NO.0
1	ANALYZER	HYDROGEN GAS LEAK DETECTOR CHECKING		REV DATE : 20/02/10
2	MAKE	MSA		REMARKS
3	TAG No	HLD 336	HLD 494	Leakage created manually by opening cylinder valve. Alarm Appeared at Unit#2 MMI and in local panel. Getting resetted in both locations after leakage arrested
4	RANGE	0 - 4 % LEL		
5	SERVICE	STORAGE PART-1	STORAGE PART-2	
6	STANDARD USED	SOURCE : FILLED H2 CYLINDER		
7	PROCESS VALUE			
8	LAST DATE OF CHECKING			
9	DATE OF CKECKING DONE			
10	DUE FOR CALIBRATION			
11	NEXT DUE DATE FOR CHECKING			
Department		O&M/C&I	EMD	OPN
Signature				
Name				
Designation				
Date				